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(54) **QUICK-RELEASE MAGNETIC CORNER
GUARD APPARATUS**

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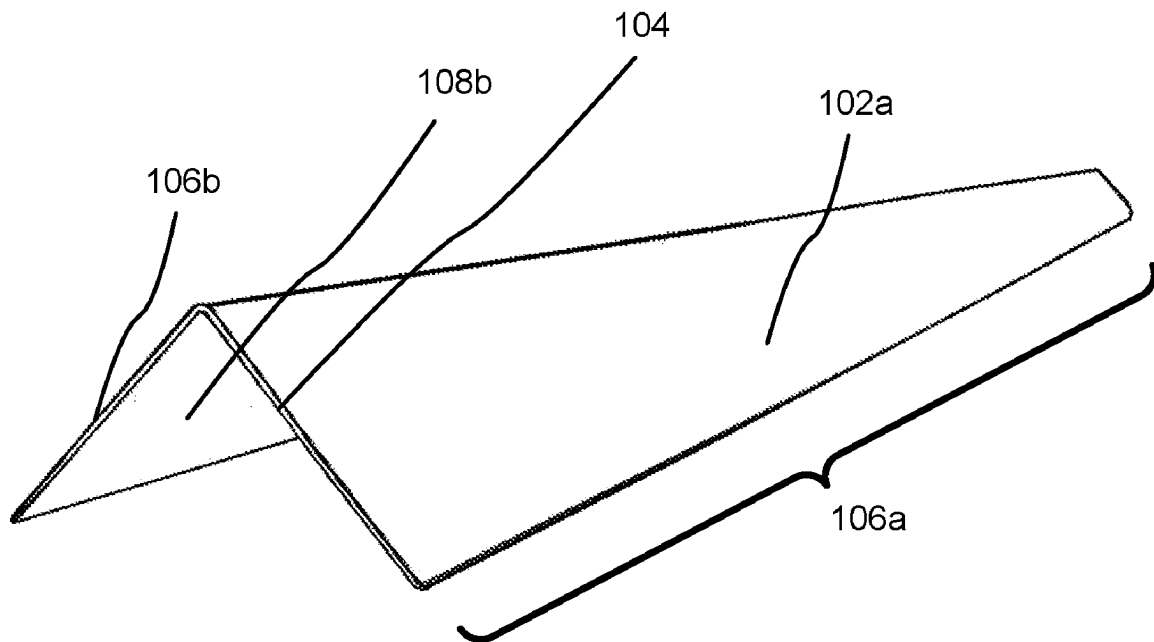
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(57) **ABSTRACT**

A corner guard having magnets operable to detachably affix the corner guard to the corner of a wall surface. In various embodiments, the corner guard comprises sidewalls hingedly connected.

100



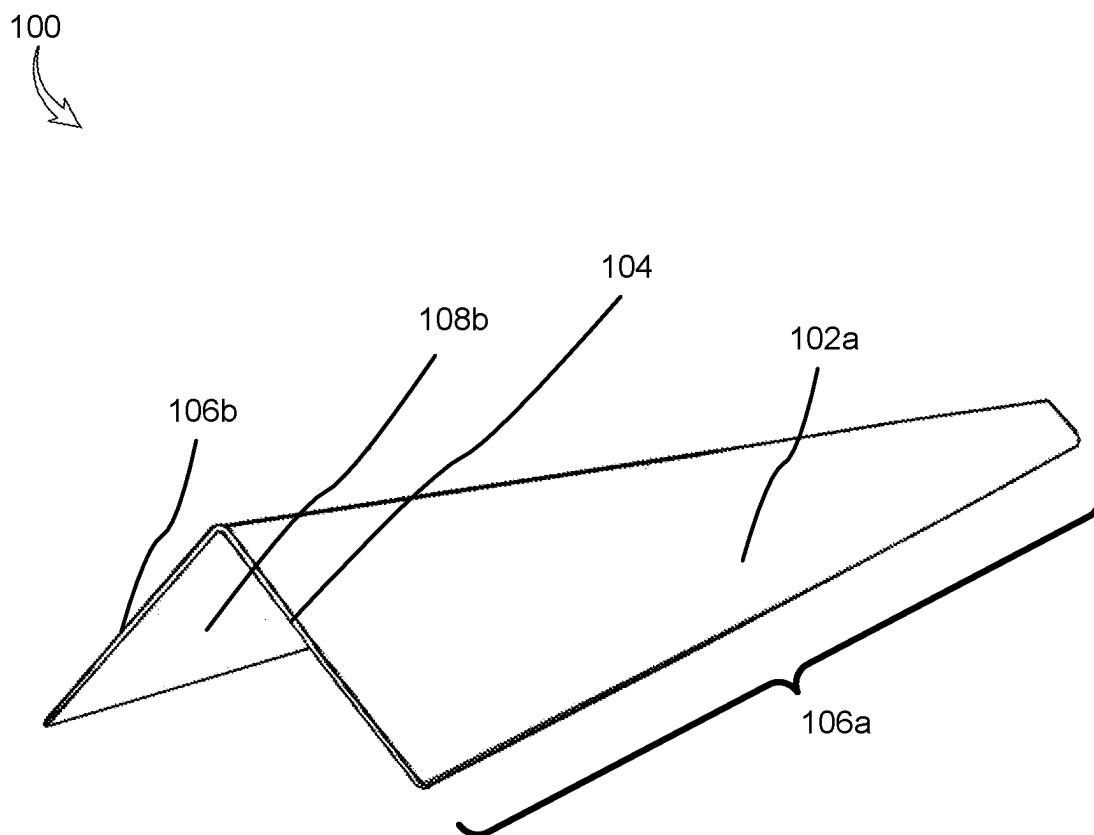


FIG. 1

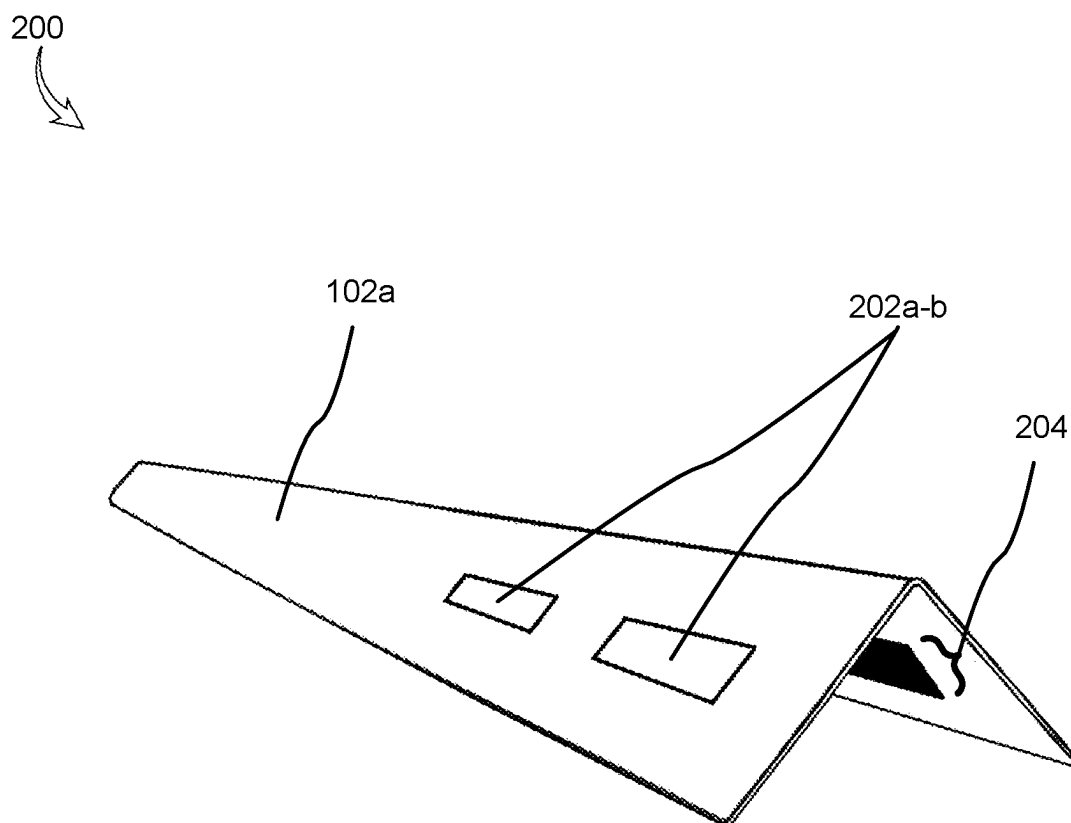


FIG. 2

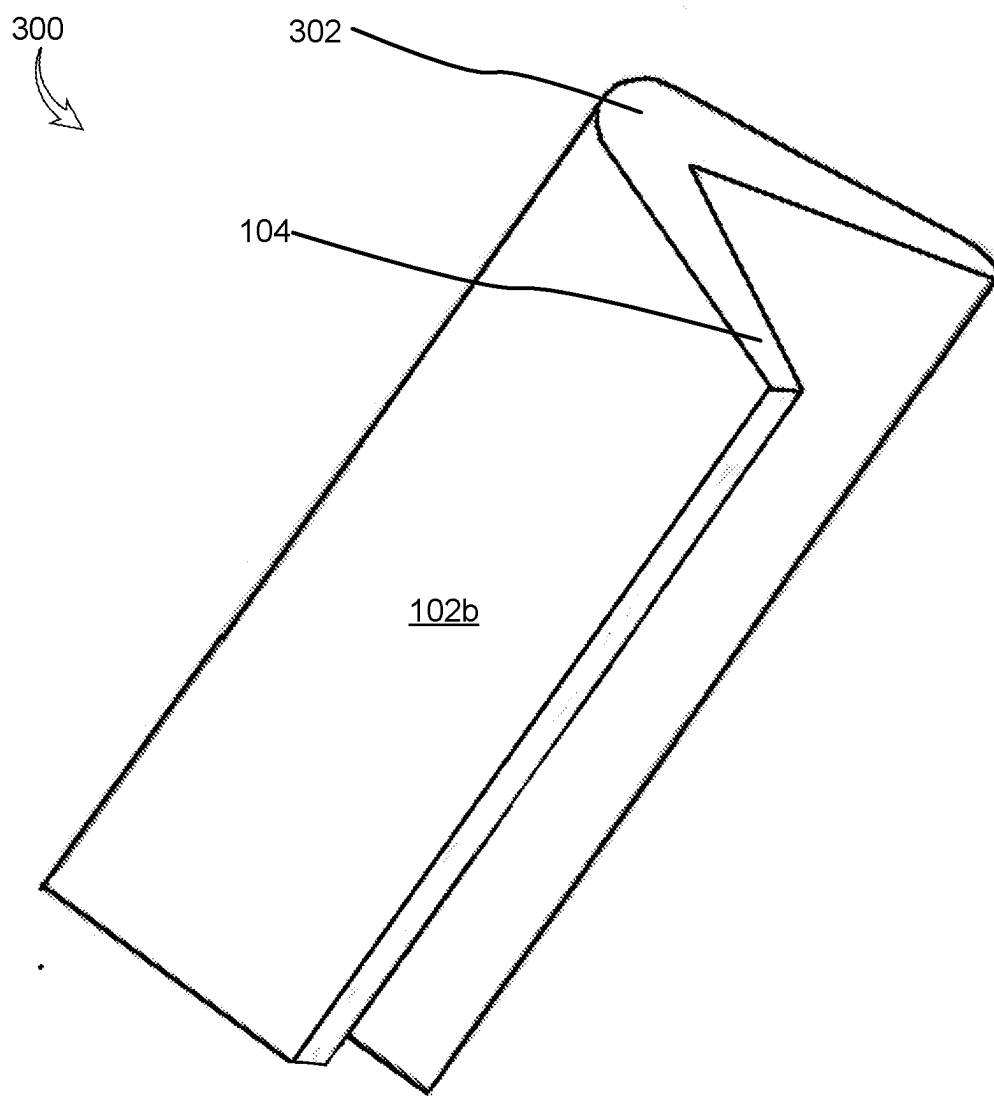


FIG. 3

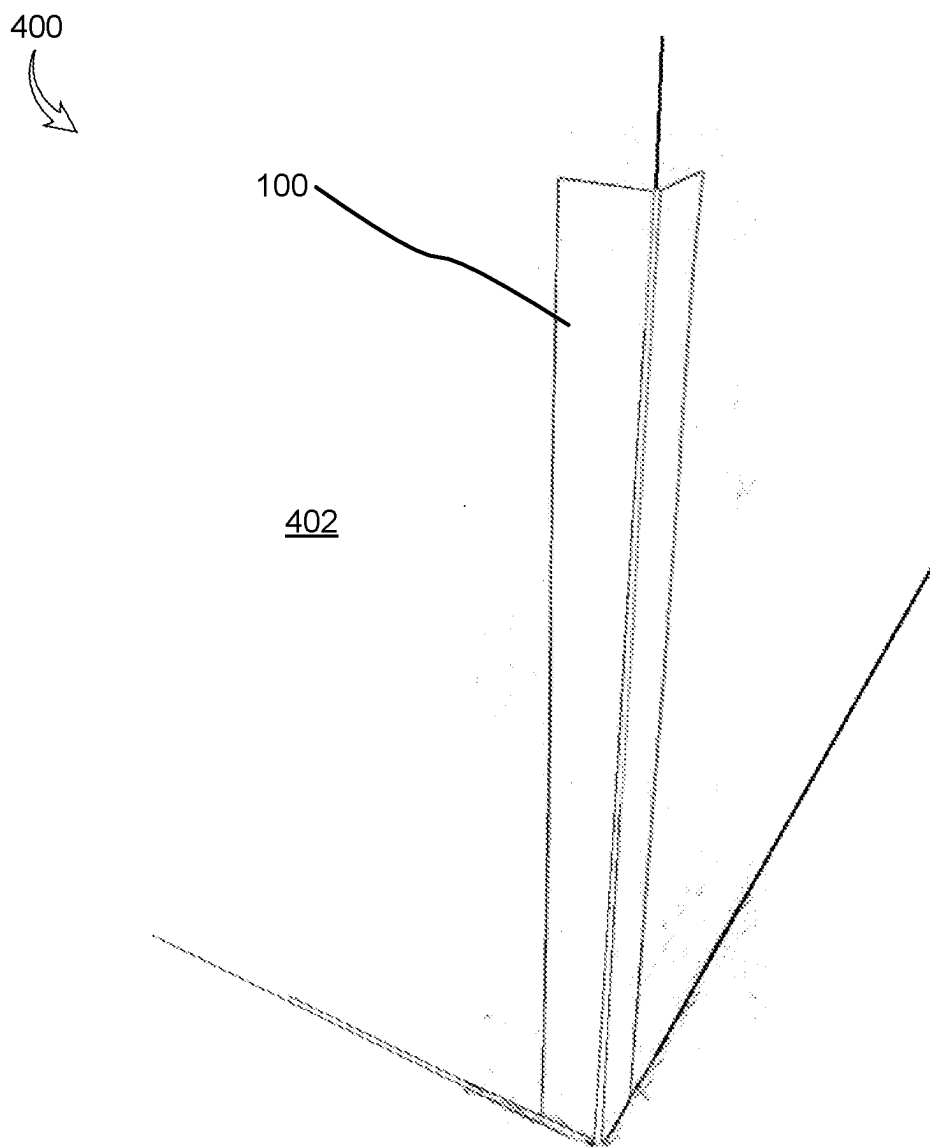


FIG. 4

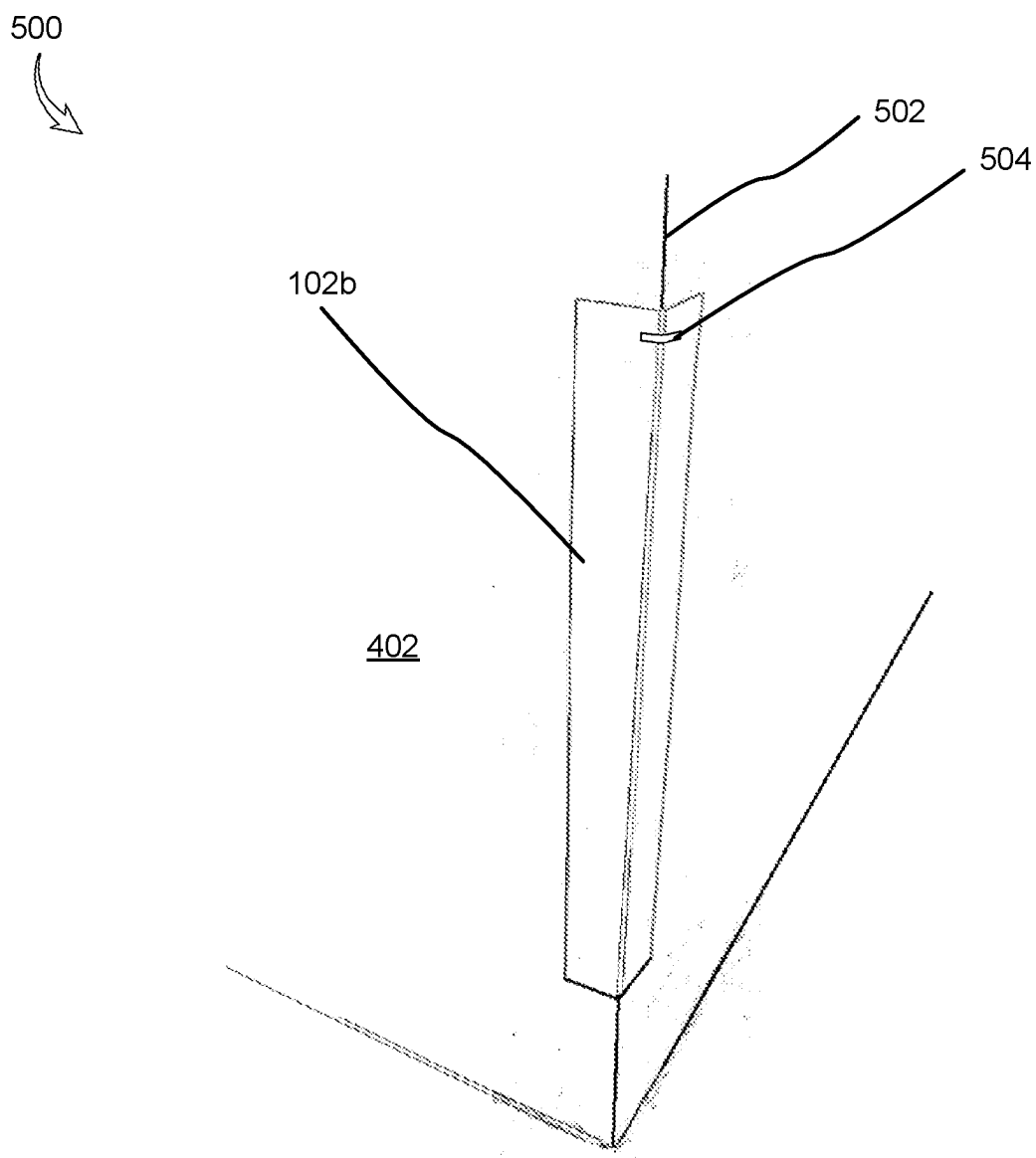


FIG. 5

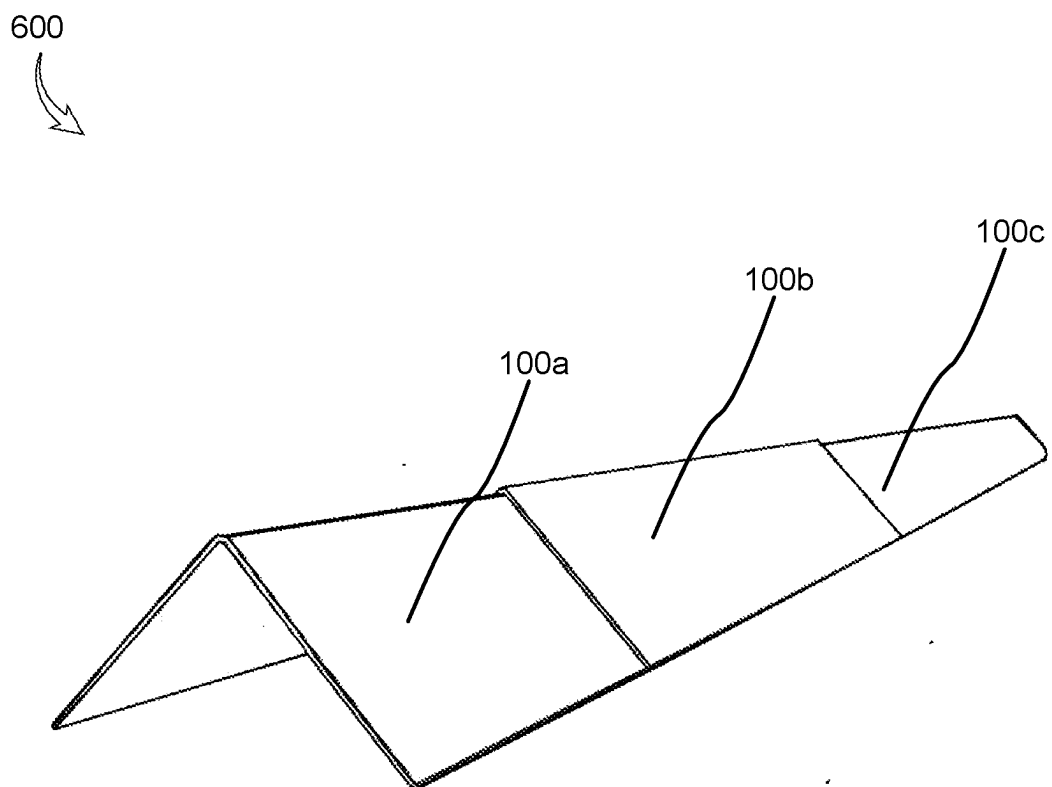


FIG. 6

700

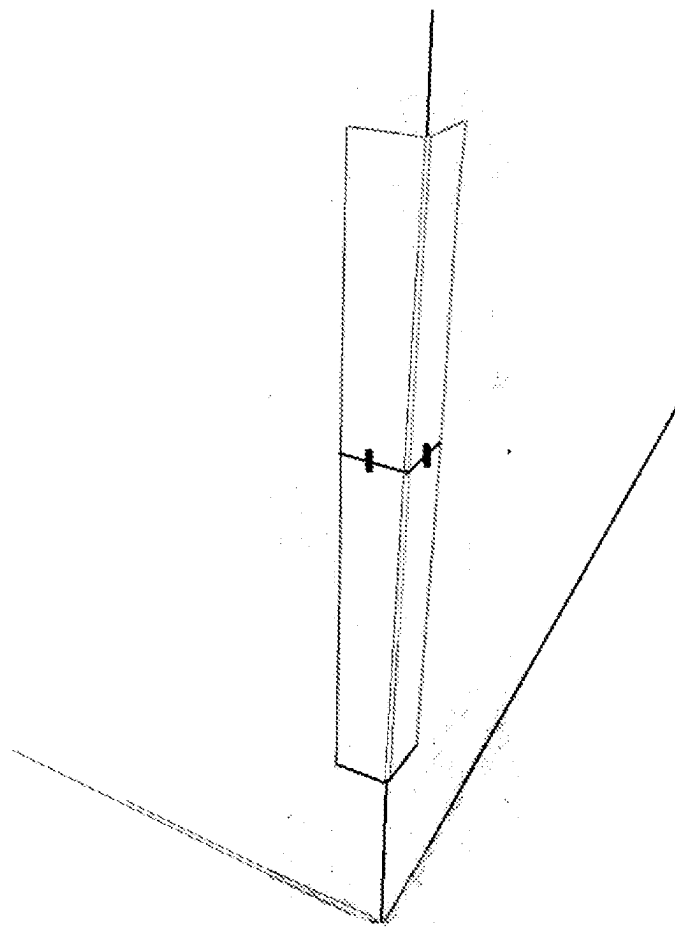



FIG. 7

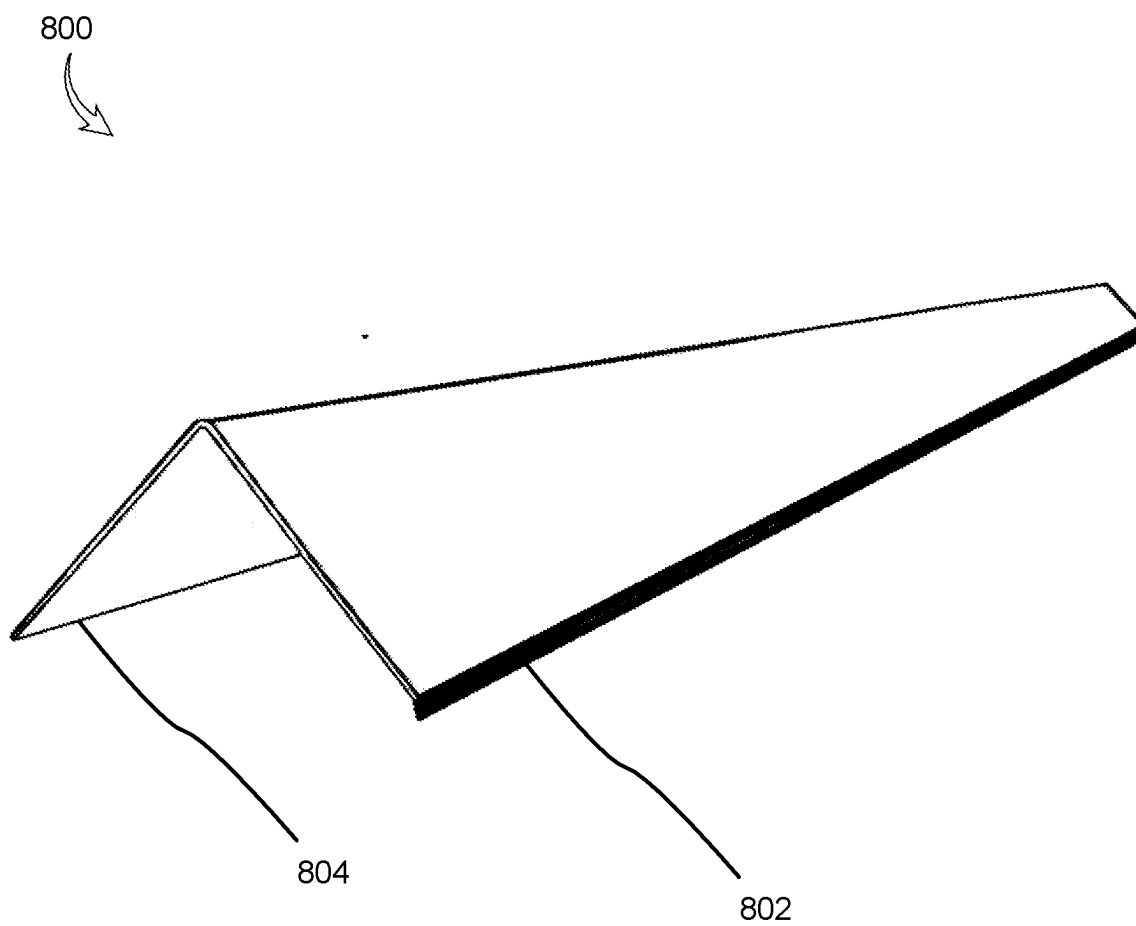


FIG. 8

QUICK-RELEASE MAGNETIC CORNER GUARD APPARATUS

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] This invention relates to safety and protective equipment, and more particularly relates to an apparatus operative to protect corners of the interior of buildings.

Description of the Related Art

[0002] Many and varied types of corner guards are known in the art, most of which must be adhered, nail or screwed into interior wall surfacing of a residential or commercial building, damaging the walls and corners which the corner guards are designed to protect. These corner guards cannot be removed with leaving holes, torn paint, damaged drywall, and the like.

[0003] Often, corner guards are only necessary during a temporary period of time during construction or moving operations, when it is desirable to place a corner guard in place which is detachable and easily to release; however, no such corner guard exists in the art.

[0004] Among the many devices which do exist are polymeric corner guards with sharp edges which often catch and damage passing furniture, building materials, dollies, and the like. There is a need in the art for an apparatus which cures these deficiencies.

SUMMARY OF THE INVENTION

[0005] From the foregoing discussion, it should be apparent that a need exists for a quick-release magnetic corner guard apparatus. Beneficially, such an apparatus would overcome many of the difficulties and inefficiencies expressed above, by providing quick-release function.

[0006] The present invention has been developed in response to the problems and needs in the art that have not yet been fully solved by currently available apparatus and methods. Accordingly, the present invention has been developed to provide a detachable corner guard, the corner guard comprising: two sidewalls affixed across lateral edges formed an angle of 87-93 degrees between said sidewalls; wherein the top surface and inner surface of the sidewalls are planar; a plurality of magnets affixed to one of an inner surface and outer surface of the sidewalls; and wherein the corner guard is longer in height than in width.

[0007] The corner guard may further comprise two edge guard comprising flexible polymeric materials, the edge guards affixed to lateral edges of the sidewalls.

[0008] The sidewalls of the corner guard may define a plurality of apertures into which the magnets are placed. The sidewalls may define a plurality of apertures into which the magnets are placed. The corner guard may further comprise an arcuate conjunction point between the lateral edges.

[0009] A second detachable corner guard is provided, the corner guard comprising: two sidewalls hingedly affixed across lateral edges to form an arch between said sidewalls; wherein the top surface and inner surface of the sidewalls are planar; a plurality of magnets affixed to one of an inner surface and outer surface of the sidewalls; and wherein the corner guard is longer in height than in width.

[0010] A detachable corner guard, the corner guard comprising: a plurality of bracket magnetically affixed to one

another, each bracket comprising: two sidewalls hingedly affixed across lateral edges to form an arch between said sidewalls; wherein the top surface and inner surface of the sidewalls are planar; a plurality of magnets affixed to one of an inner surface and outer surface of the sidewalls; and wherein the corner guard is longer in height than in width.

[0011] These features and advantages of the present invention will become more fully apparent from the following description and appended claims, or may be learned by the practice of the invention as set forth hereinafter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] In order that the advantages of the invention will be readily understood, a more particular description of the invention briefly described above will be rendered by reference to specific embodiments that are illustrated in the appended drawings. Understanding that these drawings depict only typical embodiments of the invention and are not therefore to be considered to be limiting of its scope, the invention will be described and explained with additional specificity and detail through the use of the accompanying drawings, in which:

[0013] FIG. 1 is an isometric perspective view of a quick-release magnetic corner guard apparatus in accordance with the present invention;

[0014] FIG. 2 is an isometric perspective view of a quick-release magnetic corner guard apparatus in accordance with the present invention;

[0015] FIG. 3 is a side perspective view of a quick-release magnetic corner guard apparatus in accordance with the present invention;

[0016] FIG. 4 is a forward, environmental perspective view of a quick-release magnetic corner guard apparatus in accordance with the present invention;

[0017] FIG. 5 is a forward, environmental perspective view of a quick-release magnetic corner guard apparatus in accordance with the present invention;

[0018] FIG. 6 is a side perspective view of a quick-release magnetic corner guard assembly in accordance with the present invention;

[0019] FIG. 7 is a forward, environmental perspective view of a quick-release magnetic corner guard apparatus in accordance with the present invention; and

[0020] FIG. 8 is a forward, environmental perspective view of a quick-release magnetic corner guard apparatus in accordance with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0021] Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.

[0022] Furthermore, the described features, structures, or characteristics of the invention may be combined in any suitable manner in one or more embodiments. In the following description, numerous specific details are provided to provide a thorough understanding of embodiments of the invention. One skilled in the relevant art will recognize,

however, that the invention may be practiced without one or more of the specific details, or with other methods, components, materials, and so forth. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of the invention.

[0023] It is an object of the present invention to provide a corner guard with magnetic attachment means.

[0024] FIG. 1 is an isometric perspective view of a quick-release magnetic corner guard apparatus 100 in accordance with the present invention.

[0025] The corner guard 100 comprises an L-shaped bracket having a first sidewall 106a and a second sidewall 106b. The sidewalls 106a-b are joined across a lateral edge to form an 87-93 degree angle of separation between them, or approximately a right angle.

[0026] The top surface 102a of the sidewall 106a is planar, as is the top surface of the sidewall 106b. The inner surface 108b of the sidewall 106b is also planar. The top edge 104 of the corner guard 100 is planar in the shown embodiments.

[0027] The apparatus 100 may be formed from polymeric or metal alloy materials.

[0028] FIG. 2 is an isometric perspective view of a quick-release magnetic corner guard apparatus 200 in accordance with the present invention.

[0029] The sidewall 106 may define a plurality of apertures 202a-b adapted to receive one or more magnets 204. These magnets 204 are operable to form a magnetic friction fit with metal in a wall surface 402. These magnets 204 may be square, round, ovoid, rectangular or irregularly-shaped.

[0030] FIG. 3 is a side perspective view of a quick-release magnetic corner guard apparatus in accordance 300 with the present invention.

[0031] In various embodiments, the top surfaces 102a-b may bulge outwardly, be convex, or arcuate. The conjunction 302 between the top surfaces 102a-b may be convex or arcuate.

[0032] FIG. 4 is a forward, environmental perspective view of a quick-release magnetic corner guard apparatus 400 in accordance with the present invention.

[0033] The corner guard 400 may position on the corner 502 of a wall surface 402 as shown.

[0034] FIG. 5 is a forward, environmental perspective view of a quick-release magnetic corner guard apparatus 500 in accordance with the present invention.

[0035] In various embodiments, the corner guard 500 may be suspending above a floor surface and thereby be operable to be adjustable in height.

[0036] In some embodiments, the sidewalls 106 are hingedly connected at the junction point 302 using hinges 504 such that the angle and arch between the sidewalls may be adjustable.

[0037] FIG. 6 is a side perspective view of a quick-release magnetic corner guard assembly in accordance with the present invention.

[0038] In various embodiments, a plurality of apparati 100a-c may magnetically affix to one another and/or telescope relative to one another. In this manner, the assembly 600 is operable to be adjustable in height and to fit a plurality wall sizes.

[0039] FIG. 7 is a forward, environmental perspective view of a quick-release magnetic corner guard assembly 700 in accordance with the present invention.

[0040] The assembly 700 may comprise two apparati 100a-b connected hingedly connected at edges 104.

[0041] FIG. 8 is a forward, environmental perspective view of a quick-release magnetic corner guard apparatus in accordance with the present invention.

[0042] In various embodiments, a lateral edge 804 of the sidewall 106 may comprise an edge guard 802, which may be formed from polymeric materials. The edge guard 802 is operable to protect the edge 804 from being caught on passing furniture and materials. The edge guard 802 is permanently affixed, in some embodiments, the edge 804. The edge guard curves inwardly, and may be formed from flexible material such that the edge guard 802 flexes outwardly when pressed against a wall surface 402.

[0043] The edge guard 802 may be further necessitated by the magnets 204 affixed to the inner surface 102 which cause the sidewalls 106 to protrude outwardly further than the sidewalls otherwise would in the absence of the magnets 204.

[0044] The present invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

What is claimed is:

1. A detachable corner guard, the corner guard comprising:
 - two sidewalls affixed across lateral edges formed an angle of 87-93 degrees between said sidewalls;
 - wherein the top surface and inner surface of the sidewalls are planar;
 - a plurality of magnets affixed to one of an inner surface and outer surface of the sidewalls; and
 - wherein the corner guard is longer in height than in width.
2. The corner guard of claim 1, further comprises two edge guard comprising flexible polymeric materials, the edge guards affixed to lateral edges of the sidewalls.
3. The corner guard of claim 1, wherein the sidewalls define a plurality of apertures into which the magnets are placed.
4. The corner guard of claim 1, wherein the sidewalls define a plurality of apertures into which the magnets are placed.
5. The corner guard of claim 1, further comprising an arcuate conjunction point between the lateral edges.
6. A detachable corner guard, the corner guard comprising:
 - two sidewalls hingedly affixed across lateral edges to form an arch between said sidewalls;
 - wherein the top surface and inner surface of the sidewalls are planar;
 - a plurality of magnets affixed to one of an inner surface and outer surface of the sidewalls; and
 - wherein the corner guard is longer in height than in width.
7. A detachable corner guard, the corner guard comprising:
 - a plurality of bracket magnetically affixed to one another, each bracket comprising:
 - two sidewalls hingedly affixed across lateral edges to form an arch between said sidewalls;

wherein the top surface and inner surface of the side-walls are planar;
a plurality of magnets affixed to one of an inner surface and outer surface of the sidewalls; and
wherein the corner guard is longer in height than in width.

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